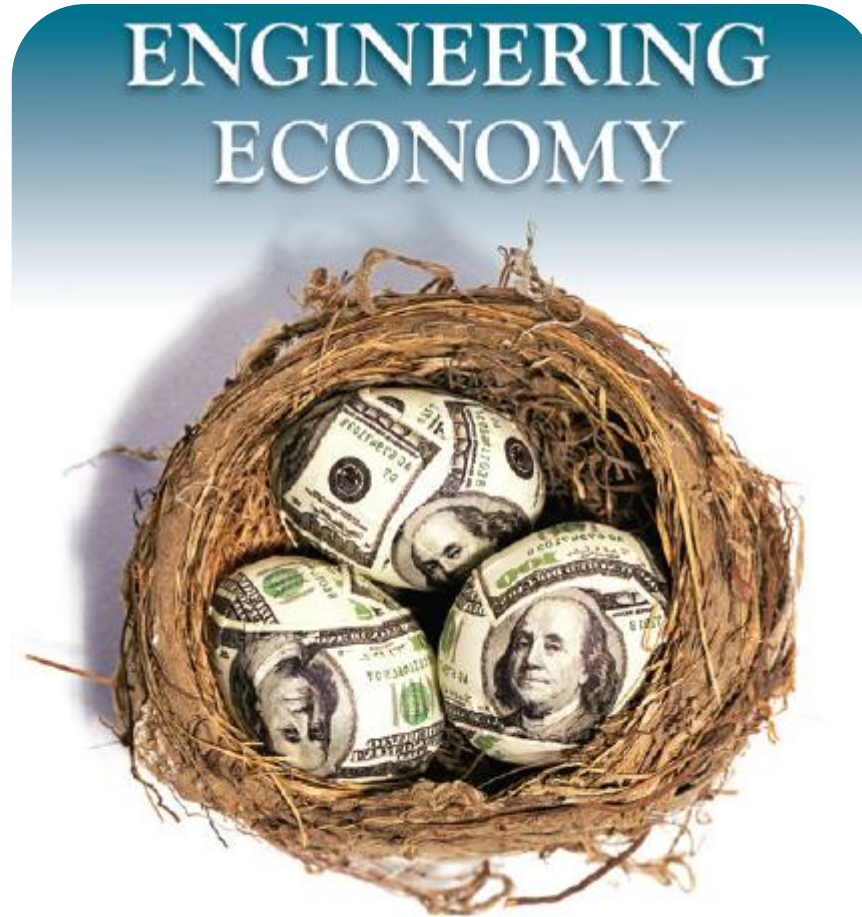




Introduction of PRE 380 Engineering Economics

Dr. Suriyaphong Nilsang

This material is modified and based on the presentation by Blank and Tarquin (2012) 7th Edition

Introduction-1

COURSE OVERVIEW

□ Course Instructors

- Dr. Suriyaphong Nilsang (Chris)

□ Contact via

- Line ID: suriyaphong.nilsang
- E-mail: suriyaphong.nils@mail.kmutt.ac.th,
suriyaphong.nilsang@gmail.com

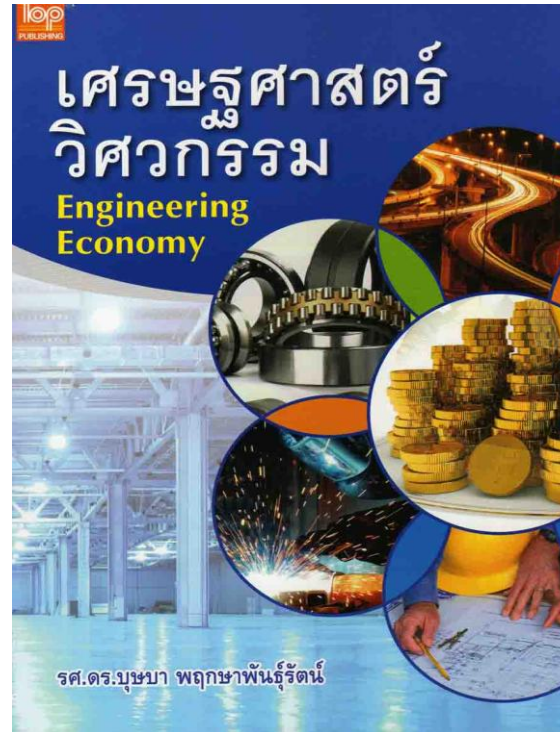
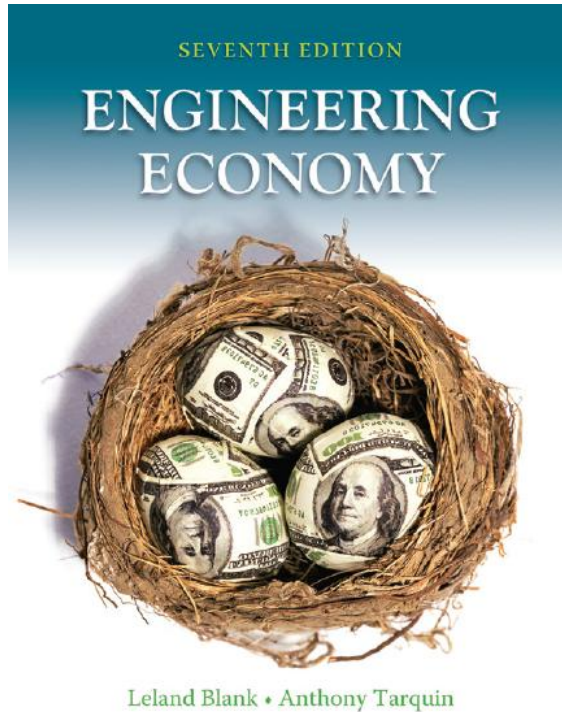
□ Office hours

- 1-4PM on Monday, Tuesday, and Friday
- or by appointment



COURSE OVERVIEW

□ Textbook



APPENDIX C

COMPOUND INTEREST TABLES

Values of Interest Factors When n Equals Infinity

Single Payment:	Uniform Payment Series:
$(F/P, i, \infty) = \infty$	$(A/F, i, \infty) = 0$
$(P/F, i, \infty) = 0$	$(A/P, i, \infty) = i$
Arithmetic Gradient Series:	$(F/A, i, \infty) = \infty$
$(A/G, i, \infty) = 1/i$	$(P/A, i, \infty) = 1/i$
$(P/G, i, \infty) = 1/i^2$	

- Compound interest table
- Engineering Economy Factors download from LEB2

COURSE OVERVIEW

□ Learning Plan

Week	Learning Activity
1	Introduction/ Foundations Of Engineering Economy
2	Factors, Effect of Time & Interest on Money
3	Combining Factors, Nominal & Effective Interest Rates
4	Present Worth Analysis
5	Annual Worth Analysis
6	ROR Analysis for a Single Alternative
7	ROR Analysis for a Multiple Alternatives
8	Midterm Examination

COURSE OVERVIEW

□ Learning Plan

Week	Learning Activity
9	Benefit & Cost Analysis
10	Making Choices: Project Financing and Noneconomic Attributes
11	Replacement and Retention Decisions
12	Breakeven Analysis
13	Effects of Inflation, Cost Estimation
14	Depreciation Methods,
15	After Tax Economic Analysis/ Decision Making under Risk
16	Final Examination

COURSE OVERVIEW

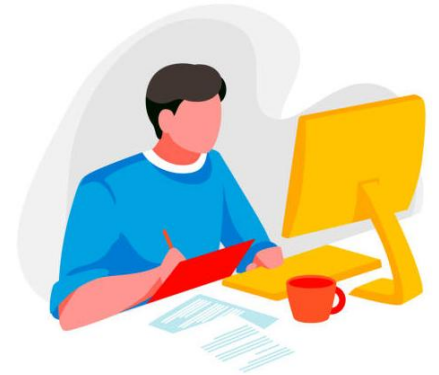
□ Learning style



**Assessments and
Namecheck in-class**



**Brief and Discuss
in class**



**Self Learning
via VIDEO**

COURSE OVERVIEW

□ Learning outcomes



Learning outcome 1

- Learners can apply the principle of the time value of money to calculate equivalent cash flows over any period or time point.



Learning outcome 2

- Learners can compare and decide on investment options by the basis tool of economic estimation.



Learning outcome 3

- Learners can analyze the Replacement study of asset, Breakeven, and Payback analysis.



Learning outcome 4

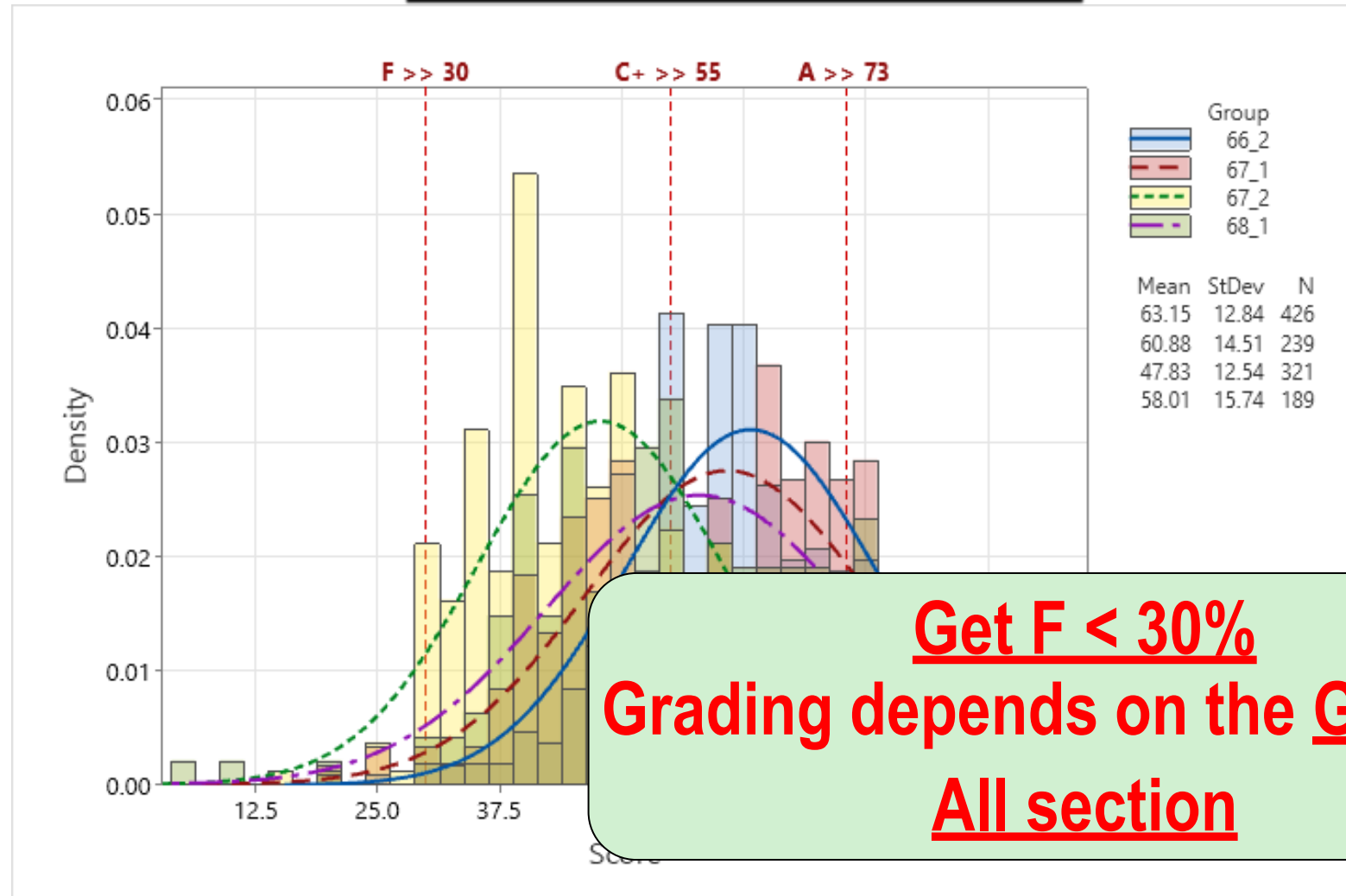
- Learners can economically compare projects under depreciation, income taxes, risk, and uncertainty.

COURSE OVERVIEW

□ Grading Policy

- | | |
|--------------------------|------------------------------|
| ➤ Midterm examination | 40 % (Score: 80/2) |
| ➤ Final examination | 40 % (Score: 80/2) |
| ➤ Quiz | 20 % (4 time: 10/2 per time) |
| ➤ Assessments (in-class) | 5 % (Extra) |
| ➤ Attendance | |

COURSE OVERVIEW



IMPORTANT DATE PRE380

Month	Week	MON	TUE	WED	THU	FRI	SAT	SUN	Plan	Quiz
ม.ค.-25	1	12	13	14	15	16	17	18	Intro/Ch.1	
ม.ค.-25	2	19	20	21	22	23	24	25		
ม.ค.-25	3	26	27	28	29	30	31	1	CH.2	
ก.พ.-25	4	2	3	4	5	6	7	8	Ch.3&4	
ก.พ.-25	5	9	10	11	12	13	14	15	Ch.5	Quiz#1
ก.พ.-25	Exam 1	16	17	18	19	20	21	22	-	
ก.พ.-25	6	23	24	25	26	27	28	1	Ch.6&7	
มี.ค.-25	7	2	3	4	5	6	7	8	Ch.8	
มี.ค.-25	8	9	10	11	12	13	14	15	Ch.9&10	Quiz#2
มี.ค.-25	9	16	17	18	19	20	21	22	Ch.11&12	
มี.ค.-25	10	23	24	25	26	27	28	29	Break	
เม.ย.-25	Exam 2	30	31	1	2	3	4	5	Midterm	
เม.ย.-25	Exam 2	6	7	8	9	10	11	12	Break	
เม.ย.-25	11	13	14	15	16	17	18	19	Break	Quiz#3
เม.ย.-25	12	20	21	22	23	24	25	26	Ch.13&14	
เม.ย.-25	13	27	28	29	30	1	2	3	Ch.14&15	
พ.ค.-25	14	4	5	6	7	8	9	10	Ch.16	
พ.ค.-25	15	11	12	13	14	15	16	17	Break	Quiz#4
พ.ค.-25	Exam 3	18	19	20	21	22	23	24	Final	
พ.ค.-25	Exam 3	25	26	27	28	29	30	31	-	

Introduction-10

Midterm exam >>

31 March 2026

- Chapter 1-9
- Onsite exam by Registration Office
- 13:00 - 16:00

Final exam >>

22 May 2026

- Chapter 10-16
- Onsite exam by Registration Office
- 13:00 - 16:00

LINE GROUP & VIDEO

Line Open chat group



LINE GROUP & VIDEO

Link VDO PRE380

A screenshot of a YouTube playlist titled "#PRE380 เศรษฐศาสตร์วิศวกรรม" (Engineering Economy) by Suriyaphong Nilsang. The playlist contains 6 videos. The first video is "Chapter 1: Foundations of Engineering Economy" with a duration of 1:11:35. The second video is "Chapter 2: Factors, Effect of Time & Interest on Money" with a duration of 48:53. The third video is "Chapter 3: Combining Factors" with a duration of 27:07. The fourth video is "Chapter 4: Nominal and Effective Interest Rates" with a duration of 37:29. The fifth video is "Chapter 5: Present Worth Analysis" with a duration of 54:49. The sixth video is "Chapter 6: Annual Worth Analysis". The playlist has 18 videos and 30,505 views.

#PRE380 เศรษฐศาสตร์วิศวกรรม

by Suriyaphong Nilsang

Playlist • Unlisted • 18 videos • 30,505 views

Play all

Manual

#PRE380 เศรษฐศาสตร์วิศวกรรม: Chapter 1: Foundations of Engineering Economy
Suriyaphong Nilsang • 2K views • 1 year ago

#PRE380 เศรษฐศาสตร์วิศวกรรม: Chapter 2: Factors, Effect of Time & Interest on Money
Suriyaphong Nilsang • 1.9K views • 1 year ago

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Suriyaphong Nilsang • 1.4K views • 1 year ago

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Suriyaphong Nilsang • 1.6K views • 1 year ago

#PRE380 เศรษฐศาสตร์วิศวกรรม: Chapter 6: Annual Worth Analysis